SG No. 52 D_{2h}^{6} $Pnna$ [orthorhombic]

* plus set: $+[0,\ 0,\ 0]$
* Wyckoff site: `4a`, site symmetry: `-1`

Table 1: Wyckoff bond: `4a@4a`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[0,\ 0,\ 0]$	`[1,-5]`
2	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[2,-6]`
3	$[-X,\ Y,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,-7]`
4	$[X,\ -Y,\ -Z]$	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[4,-8]`

* Wyckoff site: `4b`, site symmetry: `-1`

Table 2: Wyckoff bond: `4a@4b`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,-5]`
2	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[2,-6]`
3	$[-X,\ Y,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[3,-7]`
4	$[X,\ -Y,\ -Z]$	$\left[0,\ \frac{1}{2},\ 0\right]$	`[4,-8]`

* Wyckoff site: `4c`, site symmetry: `..2`

Table 3: Wyckoff bond: `4a@4c`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[\frac{1}{4},\ 0,\ z\right]$	`[1,-2]`
2	$[-X,\ Y,\ 0]$	$\left[\frac{1}{4},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[3,-4]`
3	$[-X,\ -Y,\ 0]$	$\left[\frac{3}{4},\ 0,\ -z\right]$	`[5,-6]`
4	$[X,\ -Y,\ 0]$	$\left[\frac{3}{4},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[7,-8]`

Table 4: Wyckoff bond: `4b@4c`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{1}{4},\ 0,\ z\right]$	`[1,2]`
2	$[0,\ 0,\ -Z]$	$\left[\frac{1}{4},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[3,4]`
3	$[0,\ 0,\ -Z]$	$\left[\frac{3}{4},\ 0,\ -z\right]$	`[5,6]`
4	$[0,\ 0,\ Z]$	$\left[\frac{3}{4},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[7,8]`

Table 5: Wyckoff bond: `8c@4c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{4},\ 0,\ z\right]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{4},\ 0,\ z\right]$	`[2]`
3	$[-X,\ Y,\ -Z]$	$\left[\frac{1}{4},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[3]`
4	$[X,\ -Y,\ -Z]$	$\left[\frac{1}{4},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[4]`

continued ...

Table 5

No.	vector	center	mapping
5	$[-X, -Y, -Z]$	$\left[\frac{3}{4}, 0, -z\right]$	[5]
6	$[X, Y, -Z]$	$\left[\frac{3}{4}, 0, -z\right]$	[6]
7	$[X, -Y, Z]$	$\left[\frac{3}{4}, \frac{1}{2}, z+\frac{1}{2}\right]$	[7]
8	$[-X, Y, Z]$	$\left[\frac{3}{4}, \frac{1}{2}, z+\frac{1}{2}\right]$	[8]

* Wyckoff site: 4d, site symmetry: 2..

Table 6: Wyckoff bond: 4a@4d, bond symmetry: 2..

No.	vector	center	mapping
1	$[0, Y, Z]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[1,-4]
2	$[0, -Y, Z]$	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{1}{4}\right]$	[2,-3]
3	$[0, -Y, -Z]$	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	[5,-8]
4	$[0, Y, -Z]$	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[6,-7]

Table 7: Wyckoff bond: 4b@4d, bond symmetry: 2..

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[1,4]
2	$[-X, 0, 0]$	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{1}{4}\right]$	[2,3]
3	$[-X, 0, 0]$	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	[5,8]
4	$[X, 0, 0]$	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[6,7]

Table 8: Wyckoff bond: 8c@4d, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{1}{4}\right]$	[2]
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{1}{4}\right]$	[3]
4	$[X, -Y, -Z]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[4]
5	$[-X, -Y, -Z]$	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	[5]
6	$[X, Y, -Z]$	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[6]
7	$[X, -Y, Z]$	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[7]
8	$[-X, Y, Z]$	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	[8]

* Wyckoff site: 8e, site symmetry: 1

Table 9: Wyckoff bond: 8a@8e, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}-x, -y, z\right]$	[2]
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}-x, y+\frac{1}{2}, \frac{1}{2}-z\right]$	[3]
4	$[X, -Y, -Z]$	$\left[x, \frac{1}{2}-y, \frac{1}{2}-z\right]$	[4]
5	$[-X, -Y, -Z]$	$[-x, -y, -z]$	[5]

continued ...

Table 9

No.	vector	center	mapping
6	$[X, Y, -Z]$	$\left[x+\frac{1}{2}, y, -z\right]$	[6]
7	$[X, -Y, Z]$	$\left[x+\frac{1}{2}, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[7]
8	$[-X, Y, Z]$	$\left[-x, y+\frac{1}{2}, z+\frac{1}{2}\right]$	[8]